

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5_AT2 — VISUALIZATION ANALYSIS REPORT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Evaluation (BL5)	Assessment:	Assessment Tool 2 – Visualization Analysis Report
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	<i>Evaluate the effectiveness of visualization design choices — proportional ink, axis scales, overplotting solutions, color use, and accessibility — and justify improvements. (BL5)</i>		
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Section / Batch:	B.TECH AI & DS	Date:	26.08.2026

Code of Conduct

I M.SAI HARSHITHA/192424408 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.SAI HARSHITHA

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond describing the scenario: explain WHY the design choice is flawed, and justify your judgment.
- Where a question asks for a fix, propose a specific improvement — not just “make it better.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION ANALYSIS REPORT

Rubrics for Calculation ASSESSMENT RUBRIC
AT2 — Visualization Analysis Report

CO5 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO5-AT2 Visualization Analysis Report problems, in which students critique a described visualization design choice, identify its specific flaw, and justify a well-reasoned improvement.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant visualization principle being evaluated (e.g., proportional ink, axis scale choice, overplotting, color scale design, accessibility).	Good understanding of the relevant principle, with only minor gaps.	Basic understanding, but with some misconceptions about which principle applies.	Poor understanding of core principles; misidentifies the nature of the flaw entirely.
Explanation	25%	Clearly explains why the design choice is flawed or effective, including its practical consequence for a reader — not just that it is ‘good’ or ‘bad.’	Explains the evaluation with adequate reasoning, covering most of the practical consequence.	Identifies the issue but explanation is generic or only loosely tied to its practical effect.	Little or no explanation provided; the evaluation is a bare assertion.
Application	20%	Proposes a specific, well-justified fix or recommendation that directly addresses the evaluated issue.	Proposes a workable fix, with only minor gaps in justification.	Proposes a fix, but it only partially addresses the issue or lacks justification.	Fails to propose a workable fix, or it does not address the evaluated issue.
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Organization	15%	The response is clearly structured, addressing critique and recommendation in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.
Accuracy	10%	Both the critique and the recommendation are completely accurate and well-suited to the specific scenario described.	Mostly accurate, with only minor omissions or a small error.	Partially correct; the critique or recommendation contains notable inaccuracies.	Largely incorrect or inappropriate, with major gaps in both the critique and the recommendation.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

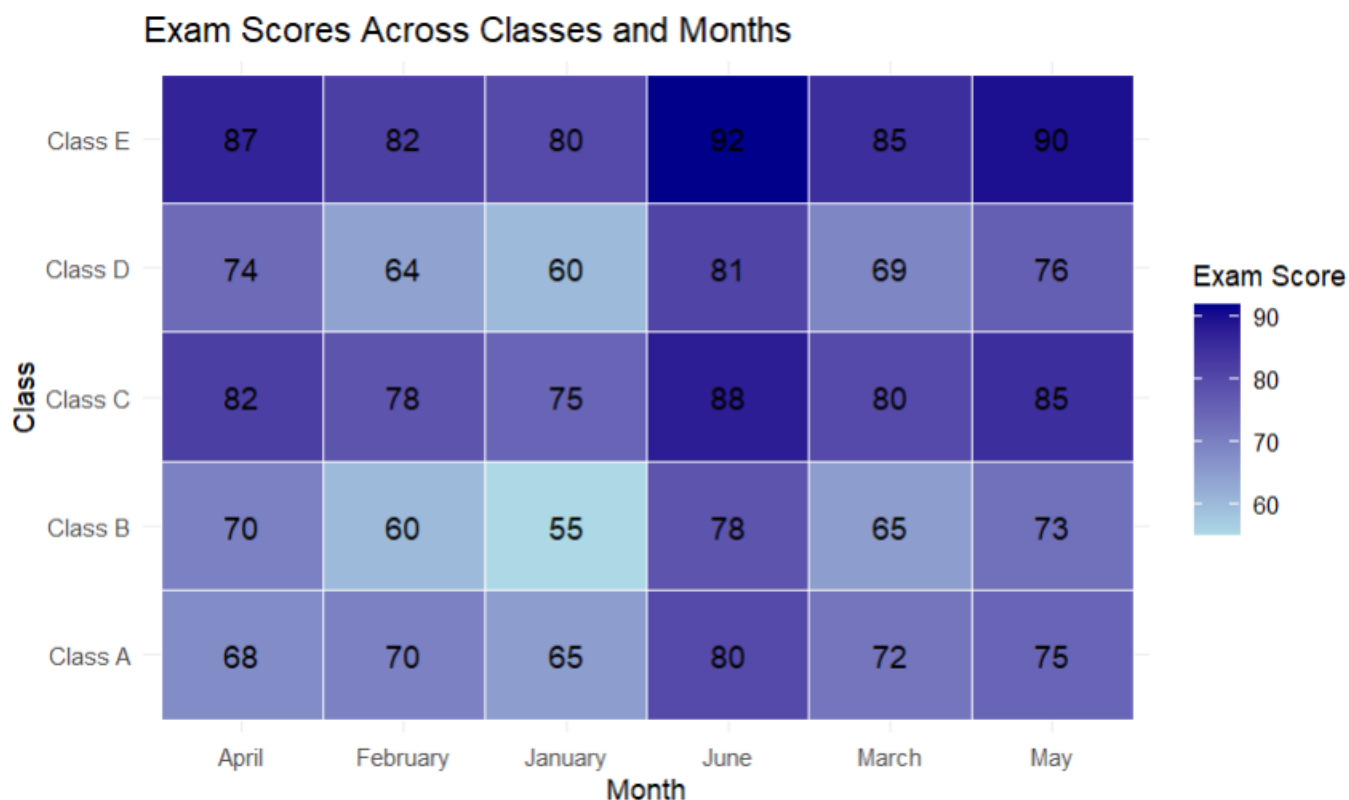
Problem 26 (5 Marks)

Data: A heatmap shows a heatmap of exam scores across 5 classes and 6 months, using a rainbow scale.

Task: Critique this color scale choice and recommend a specific alternative, explaining why it would be easier to read.

CODE:

```
library(ggplot2)
data <- data.frame(
  Class = rep(c("Class A", "Class B", "Class C", "Class D", "Class E"), each = 6),
  Month = rep(c("January", "February", "March",
    "April", "May", "June"), 5),
  Score = c(
    65, 70, 72, 68, 75, 80,
    55, 60, 65, 70, 73, 78,
    75, 78, 80, 82, 85, 88,
    60, 64, 69, 74, 76, 81,
    80, 82, 85, 87, 90, 92))
ggplot(data, aes(x = Month, y = Class, fill = Score)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Score), size = 4) +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkblue") +
  labs(
    title = "Exam Scores Across Classes and Months",
    x = "Month",
    y = "Class",
    fill = "Exam Score") +
  theme_minimal()
```



Explanation

- `geom_tile()` creates the heatmap.
- `scale_fill_gradient()` creates a **sequential blue color scale**.
- Light blue represents **lower scores**.
- Dark blue represents **higher scores**.
- `geom_text()` displays the actual score inside each cell.
- The blue scale is easier to interpret than a **rainbow scale** because it provides a clear **low-to-high progression** without distracting color changes.

CONCLUSION:

The rainbow color scale is difficult to interpret because its multiple colors can create misleading visual differences. A sequential blue color scale provides a clear light-to-dark progression from low to high exam scores. Therefore, the Blues color scale is more readable, consistent, and effective for comparing exam performance across classes and months.

Problem 27 (5 Marks) Data: A chart encodes “Pass” and “Fail” results shown only as green and red dots on a scatterplot. CO5-AT2 — Visualization Analysis Report | Student Allocation Page 12 Task: Evaluate the accessibility risk in this design for readers with color-vision deficiency, and propose a specific fix.

CODE:

```
library(ggplot2)
data <- data.frame(
  Study_Hours = c(2, 3, 4, 5, 6, 7, 8, 9, 10, 11),
  Marks = c(45, 50, 55, 60, 65, 70, 75, 80, 85, 90),
  Result = c("Fail", "Fail", "Fail", "Pass", "Pass",
             "Pass", "Pass", "Pass", "Pass", "Pass"))
ggplot(data, aes(x = Study_Hours, y = Marks,
                 shape = Result, color = Result)) +
  geom_point(size = 4) +
  scale_color_manual(
    values = c("Fail" = "#D55E00", "Pass" = "#0072B2")) +
  scale_shape_manual(
    values = c("Fail" = 4, "Pass" = 16)) +
  labs(
    title = "Student Results: Pass vs Fail",
    x = "Study Hours",
    y = "Marks",
    color = "Result",
    shape = "Result") +
  theme_minimal()
```

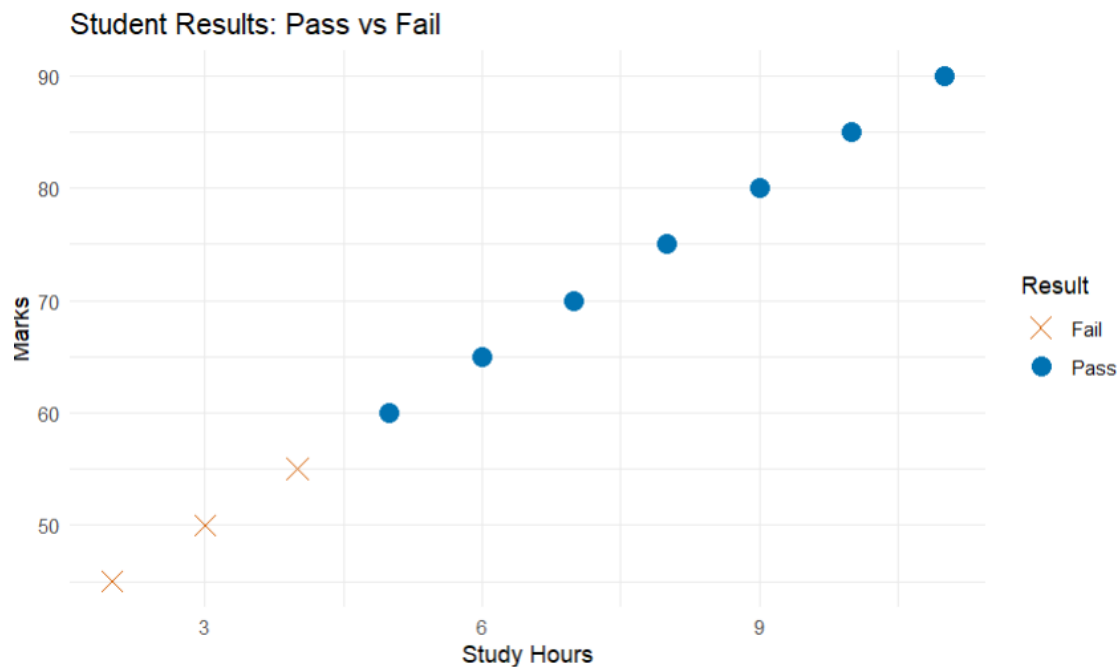
Explanation

The original scatterplot uses only **green and red dots** to represent Pass and Fail results. This creates an accessibility problem because people with **red-green color-vision deficiency** may have difficulty distinguishing these colors. As a result, they may not correctly identify whether a student has passed or failed.

To fix this problem, the visualization uses **two visual encodings: color and shape**. Pass results are represented using **blue circular points**, while Fail results are represented using **orange x symbols**. Even if a reader cannot distinguish the colors, the different shapes still clearly communicate the two categories.

Conclusion

Using only green and red is **not an accessible design** for Pass/Fail data. Combining **color with distinct shapes** makes the chart easier to understand and ensures that the information remains accessible to people with color-vision deficiency.



Problem 28 (5 Marks)

Data: A report contains two flawed charts: (1) a scatterplot of 15,000 overlapping points and (2) a bar chart with a y-axis starting at 400 for values ranging 420–460.

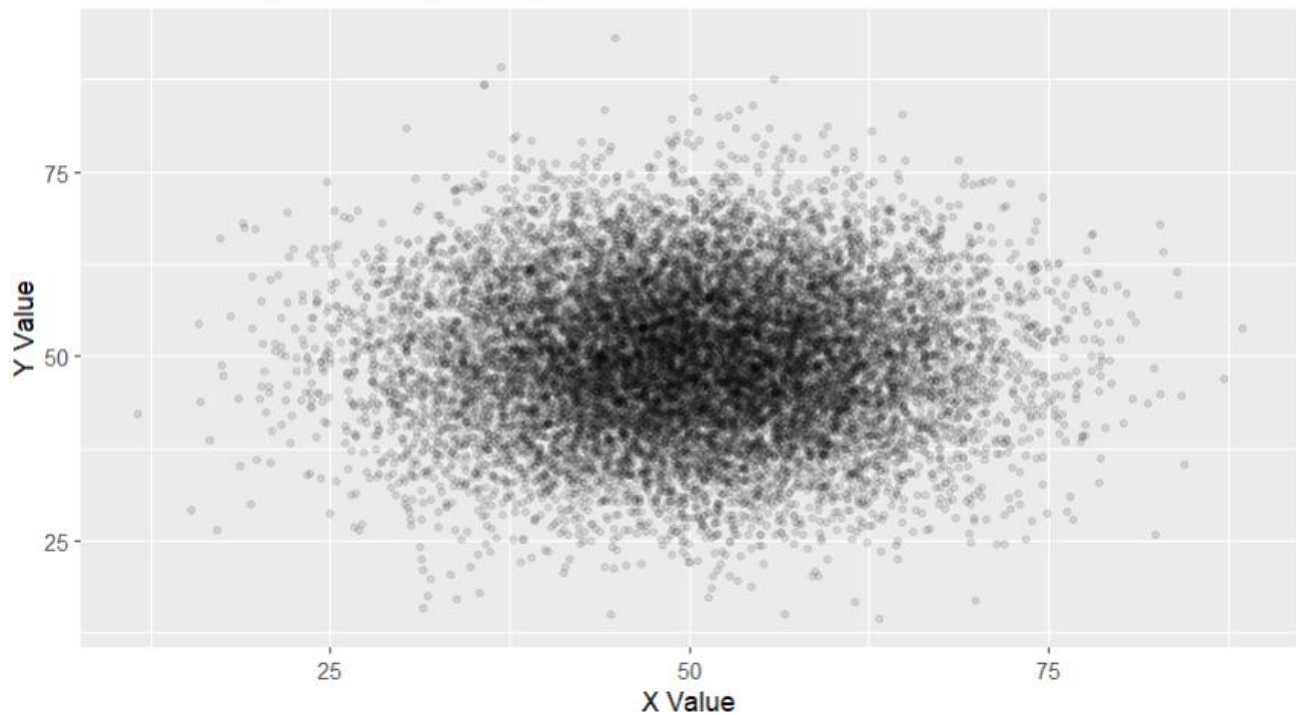
Task: Identify the specific problem in each chart, and state the fix for each.

CODE:

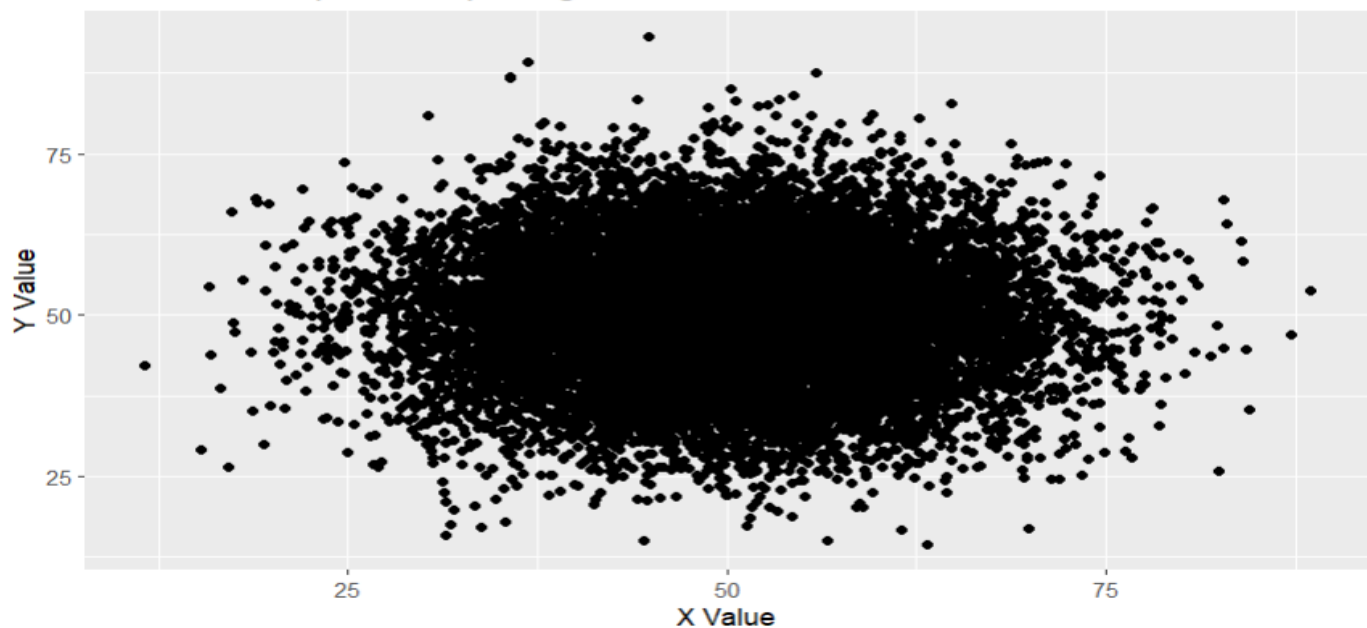
```
library(ggplot2)
set.seed(123)
data <- data.frame(
  x = rnorm(15000, mean = 50, sd = 10),
  y = rnorm(15000, mean = 50, sd = 10))
ggplot(data, aes(x, y)) +
  geom_point() +
  labs(
    title = "Flawed Scatterplot: Overplotting",
    x = "X Value",
    y = "Y Value")
ggplot(data, aes(x, y)) +
  geom_point(alpha = 0.1, size = 1) +
  labs(
    title = "Fixed Scatterplot: Transparency Used",
    x = "X Value",
    y = "Y Value")
ggplot(data, aes(x, y)) +
  geom_hex() +
  labs(
    title = "Fixed Scatterplot: Hexbin Plot",
    x = "X Value",
    y = "Y Value")
values <- data.frame(
  Category = c("A", "B", "C", "D"),
  Value = c(420, 435, 450, 460))
ggplot(values, aes(x = Category, y = Value)) +
```

```
geom_col() +
coord_cartesian(ylim = c(400, 460)) +
labs(
  title = "Flawed Bar Chart: Truncated Y-axis",
  x = "Category",
  y = "Value")
ggplot(values, aes(x = Category, y = Value)) +
geom_col() +
scale_y_continuous(limits = c(0, 500)) +
labs(
  title = "Fixed Bar Chart: Y-axis Starts at Zero",
  x = "Category",
  y = "Value")
```

Fixed Scatterplot: Transparency Used



Flawed Scatterplot: Overplotting



1. Scatterplot with 15,000 Overlapping Points

Explanation:

The original scatterplot contains **15,000 points**, causing severe **overplotting**. Many points are drawn on top of each other, so the viewer cannot clearly identify the number or density of observations.

Output:

- **Flawed output:** The scatterplot appears heavily crowded, with many points hidden behind other points.
- **Fixed output:** The transparent scatterplot clearly shows where observations are concentrated. The hexbin version shows density using different shades.

2. Bar Chart with Y-axis Starting at 400

Explanation:

The original bar chart has a **truncated y-axis**, beginning at **400 instead of 0**. Since the values range only from **420 to 460**, this makes relatively small differences look visually much larger than they really are.

Conclusion

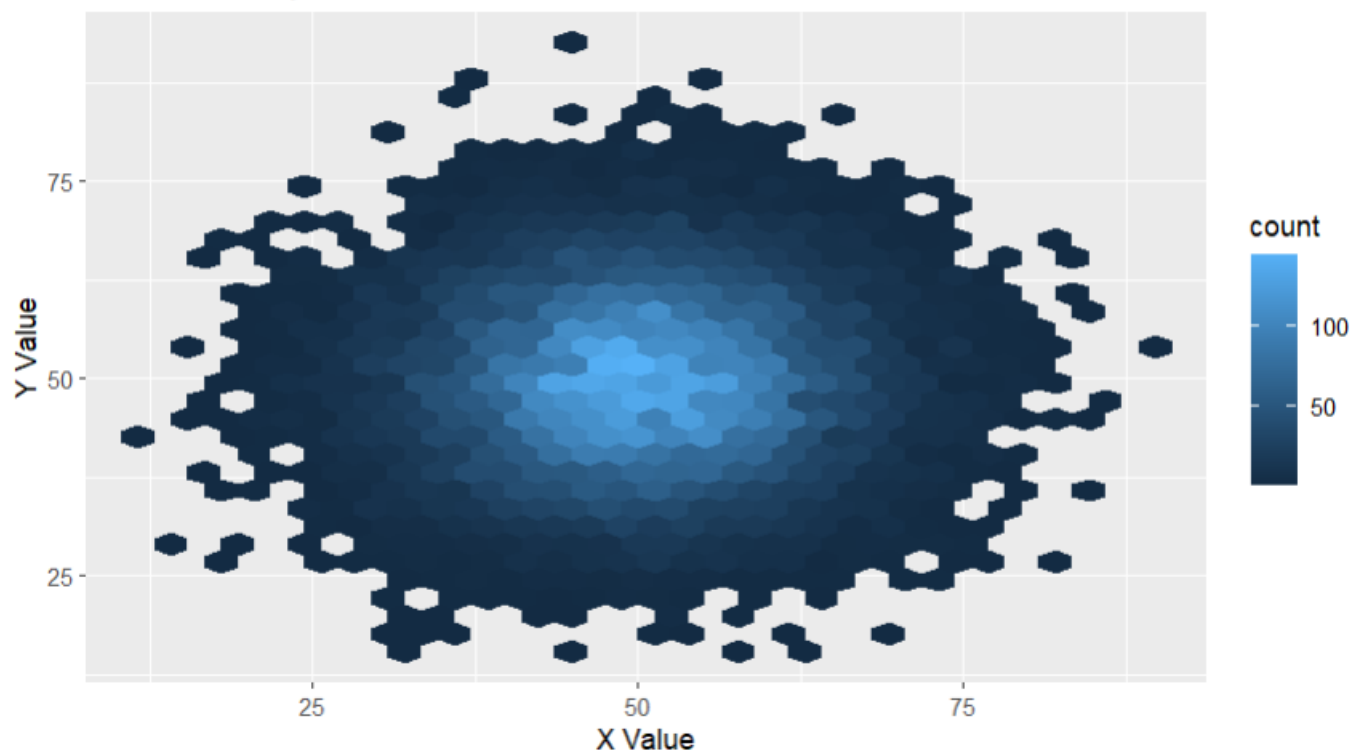
The **scatterplot problem is overplotting**, which is fixed using transparency or a hexbin plot. The **bar chart problem is a truncated y-axis**, which is fixed by starting the y-axis at **0**. These corrections make both charts clearer and prevent misleading interpretation.

Problem 29 (5 Marks)

Data: A bar chart shows weekly attendance of 910, 930, 945 students, plotted on a y-axis ranging 880–950.

Task: Explain how this axis choice violates the principle of proportional ink, and state the specific fix.

Fixed Scatterplot: Hexbin Plot



CODE:

```
attendance <- c(910, 930, 945)
weeks <- c("Week 1", "Week 2", "Week 3")
plot(1:3,
     attendance,
```



```

type = "n",
ylim = c(880, 950),
xlim = c(0.5, 3.5),
xaxt = "n",
xlab = "Week",
ylab = "Number of Students",
main = "Flawed Bar Chart")
axis(1, at = 1:3, labels = weeks)
rect(
  xleft = 1:3 - 0.3,
  ybottom = 880,
  xright = 1:3 + 0.3,
  ytop = attendance)
plot(1:3,
  attendance,
  type = "n",
  ylim = c(0, 1000),
  xlim = c(0.5, 3.5),
  yaxt = "n",
  xlab = "Week",
  ylab = "Number of Students",
  main = "Corrected Bar Chart")
axis(1, at = 1:3, labels = weeks)
rect( xleft = 1:3 - 0.3,
  ybottom = 0,
  xright = 1:3 + 0.3,
  ytop = attendance)

```

Explanation

The bar chart shows attendance values of **910, 930, and 945 students**, but the y-axis starts at **880** instead of **0** and ends at **950**.

This violates the **principle of proportional ink** because the visual length of each bar does not represent its actual value proportionally. The bars are drawn only for the small range from **880–950**, making the differences between 910, 930, and 945 appear much larger than they actually are. For example, the difference between **910 and 945 is only 35 students**, but the truncated axis can make this difference look visually dramatic.

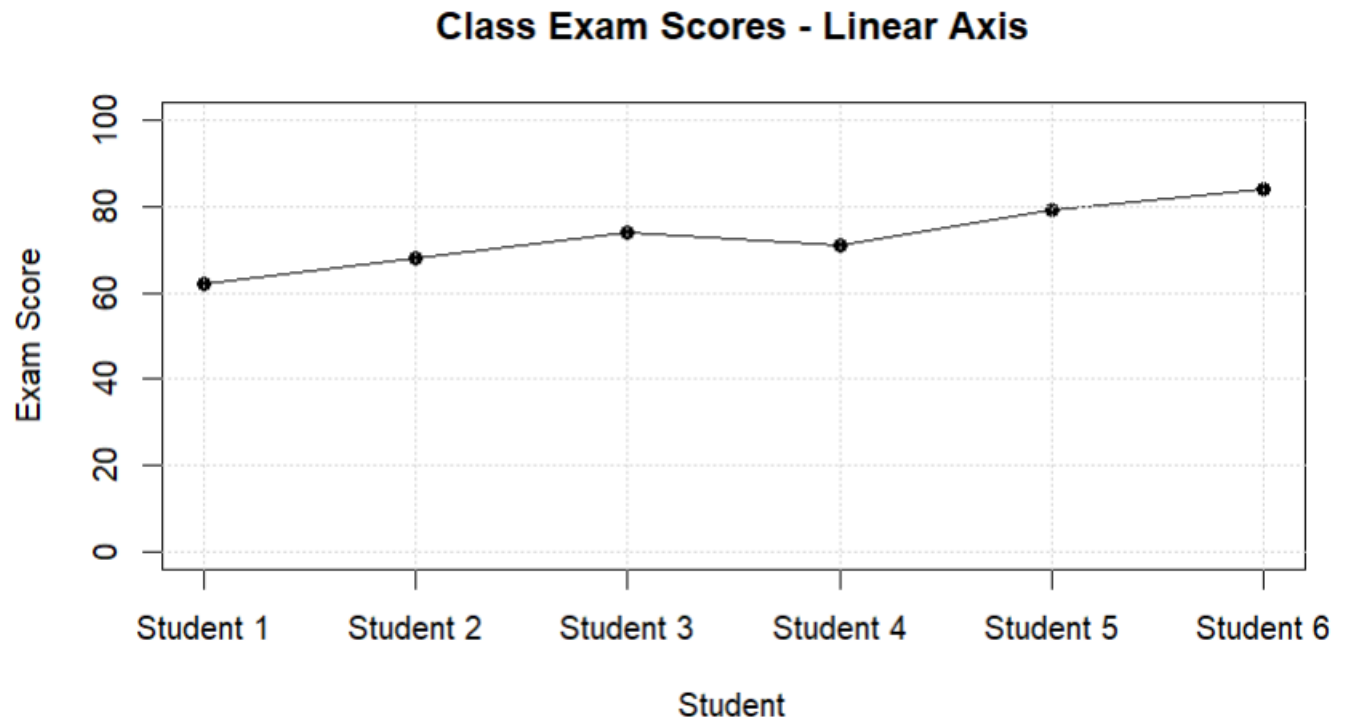
Conclusion

The **880–950 truncated y-axis exaggerates differences** and violates proportional representation of bar lengths. The specific fix is to **start the y-axis at 0**.

Problem 30 (5 Marks)

Data: Data: a class's exam scores: 62, 68, 74, 71, 79, 84.

Task: Explain why a linear axis is the appropriate choice for this data, referencing its range.



CODE:

```
scores <- c(62, 68, 74, 71, 79, 84)
minimum <- min(scores)
maximum <- max(scores)
score_range <- maximum - minimum
print(paste("Minimum Score:", minimum))
print(paste("Maximum Score:", maximum))
print(paste("Range:", score_range))
plot( scores,
      type = "o",
      pch = 19,
      xaxt = "n",
      xlab = "Student",
      ylab = "Exam Score",
      main = "Class Exam Scores - Linear Axis",
      ylim = c(0, 100))
axis(1,
     at = 1:6,
     labels = paste("Student", 1:6))
grid()
```

Explanation

The exam scores are:

62, 68, 74, 71, 79, 84

The **minimum score is 62** and the **maximum score is 84**, giving a range of:

$84 - 62 = 22$ marks

A **linear axis** is appropriate because the data has a relatively small and continuous range of **62–84**. A linear scale represents equal differences with equal visual distances. For example, the difference between 62 and 68 is 6 marks, and this difference will be shown proportionally on the axis.

A logarithmic or other nonlinear axis is unnecessary because the scores are not spread across several orders of magnitude.

Conclusion:

Since the scores range from **62 to 84 (range = 22)** and represent ordinary numerical measurements, a **linear axis is the most appropriate choice**. It provides an accurate and easy-to-interpret representation of the differences between scores.